$$\begin{aligned}
\text{Maximize} \quad & c_1x_1 + c_2x_2 + c_3x_3 + c_4x_4 \\ & \quad + c_5x_5 + ... + c_mx_m \mathit{gottamakethislonger} && (1) \\
\text{s.t.} \quad & a_{i1}x_1 + a_{i2}x_2 + a_{i3}x_3 \\ & \quad + a_{i4}x_4 + ... + a_{im}x_nm \leq b_i \qquad \forall\ i \in [\![n]\!] && (2) \\
& \mathbf{x} \geq \mathbf{0} && (3)
\end{aligned}$$

$$\underset{x\in\mathbb{R}^n}{\text{minimize}} \qquad c_1x_2 + c_2x_2 + \cdots + c_nx_n \tag{LP.a}$$

$$\begin{aligned}
\text{s.t.} \qquad & \mathbf{a}_i^\top \mathbf{x} \ \geq\ b_i && \forall\ i \in [\![m]\!] && \text{(LP.b)} \\
& \sum_{j=1}^{m} a_{ij}x_j \ \geq\ b_i && \forall\ i \in [\![m]\!] && \text{(LP.c)} \\
& \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \\ a_{m1} & a_{m2} & & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \ \geq\ \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} && && \text{(LP.d)} \\
& a_{i1}x_1 + a_{i2}x_2 + a_{i3}x_3 \\ & \quad + a_{i4}x_4 + ... + a_{im}x_nm \ \leq\ b_i && \forall\ i \in [\![n]\!] && \text{(LP.e)} \\
& && \mathbf{x} \in \mathbb{R}^n && \text{(LP.f)}
\end{aligned}$$

$$\begin{array}{rl}
\max & c^\top x \\
\text{s.t.} & Ax + s = b + 0 \\
& x, s \geq 0
\end{array}$$